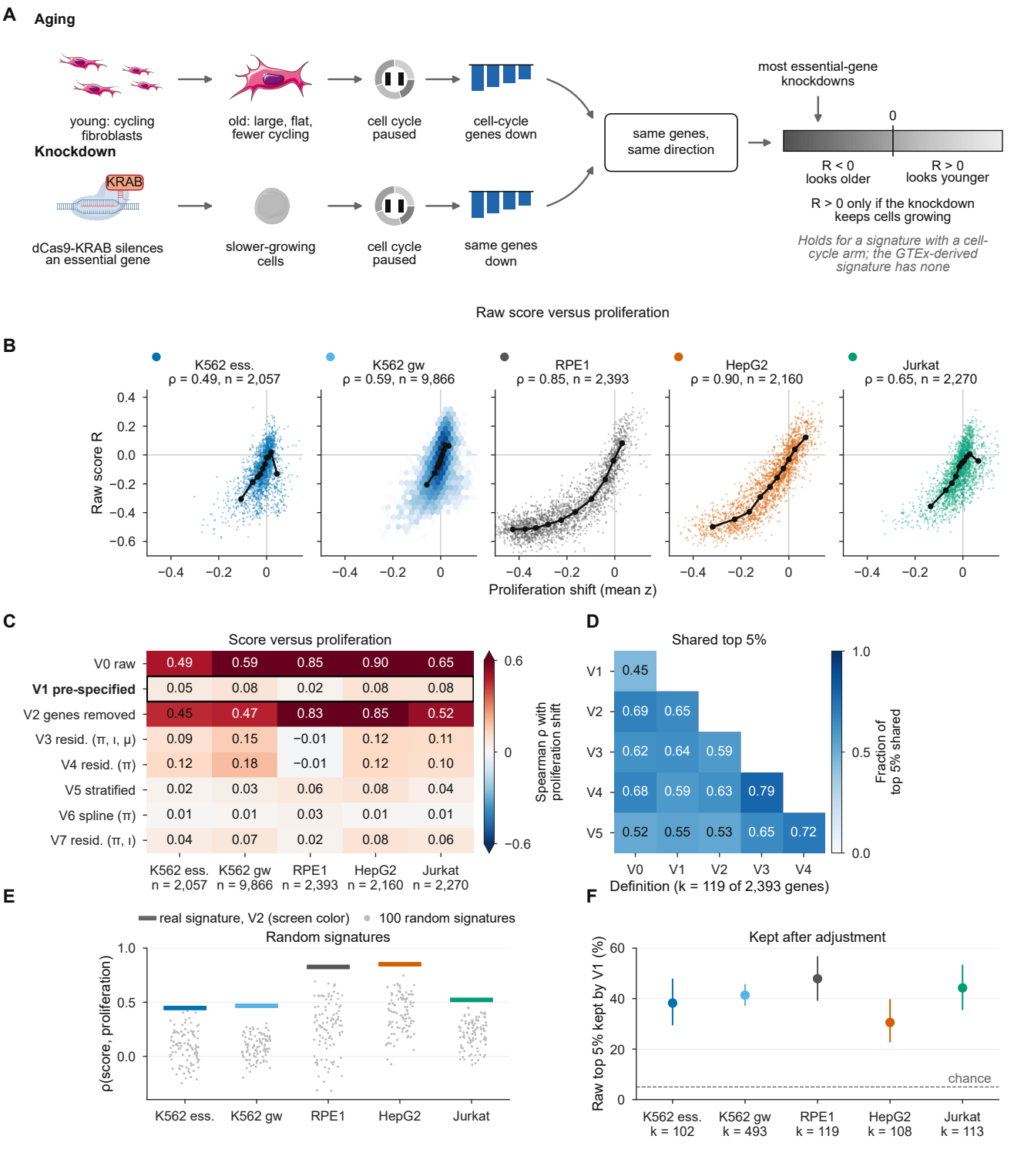

A
Aging
young: cycling fibroblasts
old: large, flat, fewer cycling
cell cycle paused
cell-cycle genes down
Knockdown
KRAB
dCas9-KRAB silences an essential gene
slower-growing cells
cell cycle paused
same genes down
same genes, same direction
most essential-gene knockdowns
0
R < 0 looks older
R > 0 looks younger
R > 0 only if the knockdown keeps cells growing
Holds for a signature with a cell-cycle arm; the GTEx-derived signature has none
B
Raw score versus proliferation
K562 ess. ρ = 0.49, n = 2,057
K562 gw ρ = 0.59, n = 9,866
RPE1 ρ = 0.85, n = 2,393
HepG2 ρ = 0.90, n = 2,160
Jurkat ρ = 0.65, n = 2,270
Raw score R
Proliferation shift (mean z)
C
Score versus proliferation
V0 raw 0.49 0.59 0.85 0.90 0.65
V1 pre-specified 0.05 0.08 0.02 0.08 0.08
V2 genes removed 0.45 0.47 0.83 0.85 0.52
V3 resid. (π, ι, μ) 0.09 0.15 −0.01 0.12 0.11
V4 resid. (π) 0.12 0.18 −0.01 0.12 0.10
V5 stratified 0.02 0.03 0.06 0.08 0.04
V6 spline (π) 0.01 0.01 0.03 0.01 0.01
V7 resid. (π, ι) 0.04 0.07 0.02 0.08 0.06
K562 ess. n = 2,057
K562 gw n = 9,866
RPE1 n = 2,393
HepG2 n = 2,160
Jurkat n = 2,270
Spearman ρ with proliferation shift
D
Shared top 5%
V1 0.45
V2 0.69 0.65
V3 0.62 0.64 0.59
V4 0.68 0.59 0.63 0.79
V5 0.52 0.55 0.53 0.65 0.72
Fraction of top 5% shared
Definition (k = 119 of 2,393 genes)
E
real signature, V2 (screen color)
100 random signatures
Random signatures
ρ(score, proliferation)
K562 ess.
K562 gw
RPE1
HepG2
Jurkat
F
Kept after adjustment
Raw top 5% kept by V1 (%)
chance
K562 ess. k = 102
K562 gw k = 493
RPE1 k = 119
HepG2 k = 108
Jurkat k = 113